

Name: _____ Date: _____

Period: _____

Primary Objective

By the end of this lesson, I will be able to...

1. Rewrite a polynomial or rational expression in _____ form or _____ form and explain what each form reveals. (LO 1.11.A)
2. Perform polynomial _____ to write $f(x) = g(x)q(x) + r(x)$ and use the result to find _____ asymptotes. (LO 1.11.B)
3. Apply the _____ theorem and Pascal's Triangle to expand $(a + b)^n$. (LO 1.11.C)

Vocabulary & Key Concepts**factored form:**

standard form:

polynomial long division:

quotient:

remainder:

slant asymptote:

binomial theorem:

Pascal's Triangle:

Hook

Three rational functions are written on the board:

$$f(x) = \frac{2x^2 - 8}{x^2 + x - 6} \quad g(x) = \frac{2(x-2)(x+2)}{(x-2)(x+3)} \quad h(x) = \frac{2(x+2)}{x+3}, \quad x \neq 2$$

These are the *same function* in three different faces. _____ form reveals zeros and holes. _____ form reveals end behavior. The simplified form reveals the _____ of the restricted domain.

LO 1.11.A — Factored Form vs. Standard Form

	Factored Form	Standard Form
What it reveals	_____	_____
	zeros, holes, asymptotes	end behavior, degree, leading coefficient
Example (poly)	$(x-1)(x+3)(x-2)$	$x^3 - 7x + 6$ (expand)
Example (rational)	$\frac{(x-2)(x+2)}{(x-2)(x+3)}$	$\frac{x^2 - 4}{x^2 + x - 6}$
Key idea (EK 1.11.A.1):	Factored form reveals _____, _____, _____, and domain restrictions.	
Key idea (EK 1.11.A.2):	Standard form reveals _____.	

LO 1.11.B — Polynomial Long Division

Division Algorithm: If f is divided by g (with $\deg g \geq 1$), then

$$f(x) = g(x) \cdot q(x) + r(x), \quad \text{where } \deg(r) < \deg(g).$$

Steps:

- Write numerator and denominator in _____ order (fill gaps with 0 coefficients).
- Divide the _____ term of the dividend by the leading term of the divisor. Write this as the first term of $q(x)$.
- Multiply that term by the _____ and subtract from the dividend.
- Bring down the next term and repeat until $\deg(\text{remainder}) < \deg(\text{divisor})$.

Example: Divide $f(x) = 2x^2 + 3x - 5$ by $g(x) = x - 1$.

$$\begin{array}{r}
 x-1 \overline{) 2x^2 + 3x - 5} \\
 \underline{-(2x^2 - 2x)} \\
 5x - 5 \\
 \underline{-(5x - 5)} \\
 0
 \end{array}$$

Result: $q(x) =$ _____, $r(x) =$ _____, so $f(x) = (x-1)(\text{_____}) +$ _____.

Slant Asymptote (EK 1.11.B.2): When $\deg(f) = \deg(g) + 1$, the slant asymptote of $\frac{f(x)}{g(x)}$ is $y = q(x)$ (the _____ after long division).

LO 1.11.C — Binomial Theorem and Pascal's Triangle

Pascal's Triangle (fill in the missing entries):

				1			
			1		1		
		1		2		1	
	1		3		3		1
1		_____		_____		_____	1

Rule: Each entry equals the _____ of the two entries above it.

Binomial Theorem (EK 1.11.C.1):

$$(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

The coefficients $\binom{n}{k}$ match row n of Pascal's Triangle.

Example: Expand $(x + 2)^3$.

Using row 3 of Pascal's Triangle (1, 3, 3, 1):

$$(x + 2)^3 = 1 \cdot x^3 + 3 \cdot x^2(\text{_____}) + 3 \cdot x(\text{_____}) + 1 \cdot (\text{_____}) = \text{_____}$$